Mediation & moderation

Readings for today

- Preacher, K. J., & Hayes, A. F. (2008). Asymptotic and resampling strategies for assessing and comparing indirect effects in multiple mediator models. Behavior research methods, 40(3), 879-891.

Topics

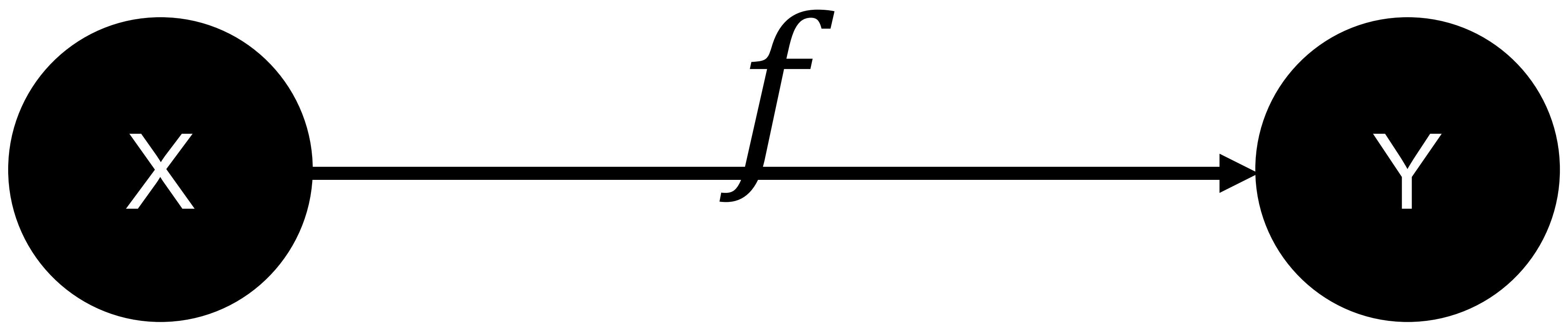
1. Graphs
2. Moderation
3. Mediation

Graphs

Graphs

Statistical model: $f(X) = Y$

Graphical form:

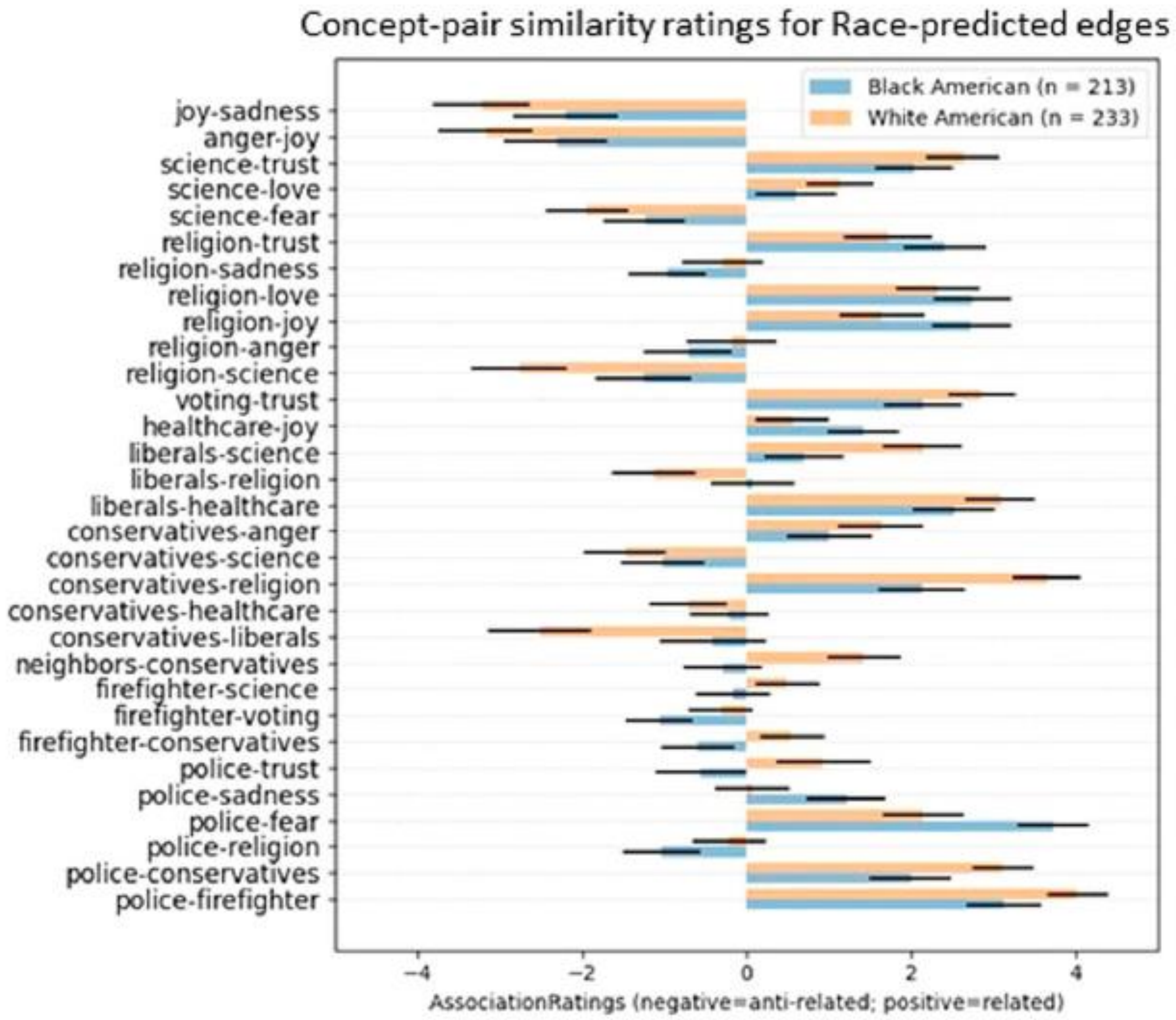


Nodes: The objects (variables)

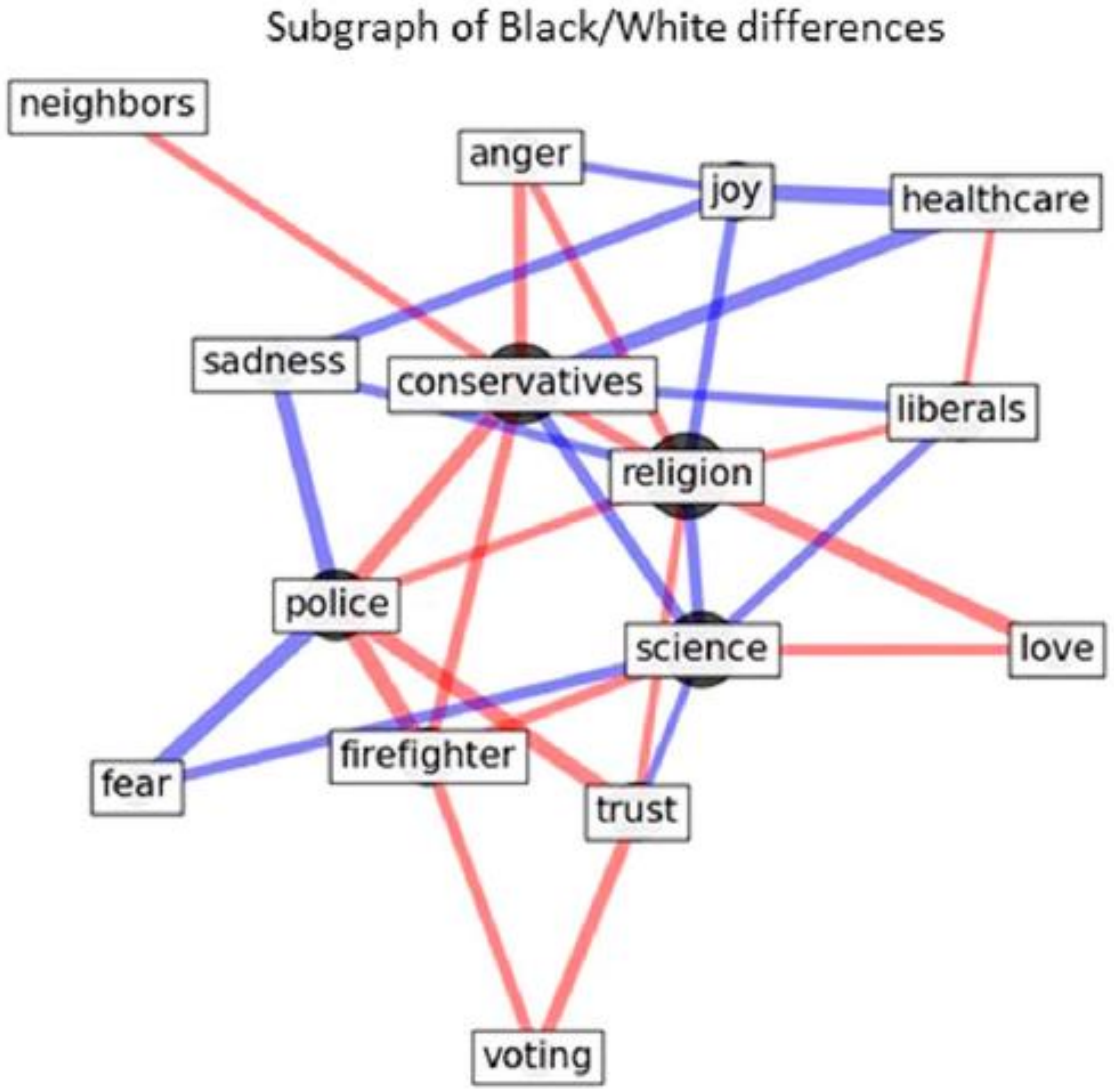
Edges: The connections (relations)

Graphs

Bar form



Graphical form

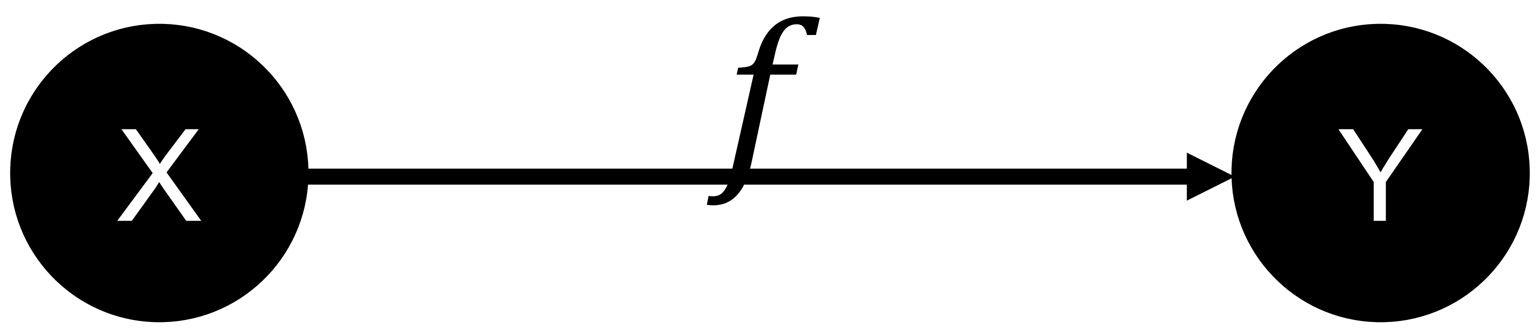


(Vargas & Verstynen, 2025)

Types of graphs

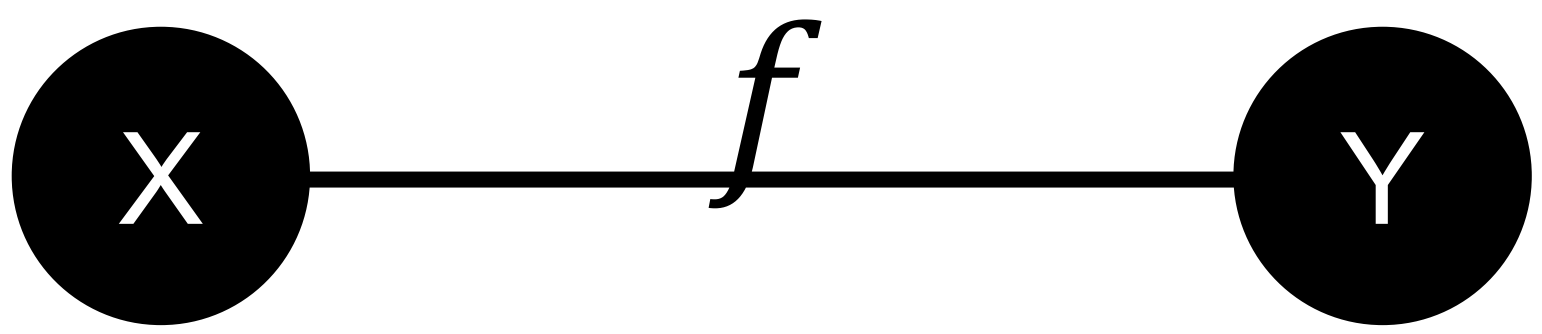
<u>Directed graphs:</u>

- “causal”
- regression



<u>Undirected graphs:</u>

- association
- correlation



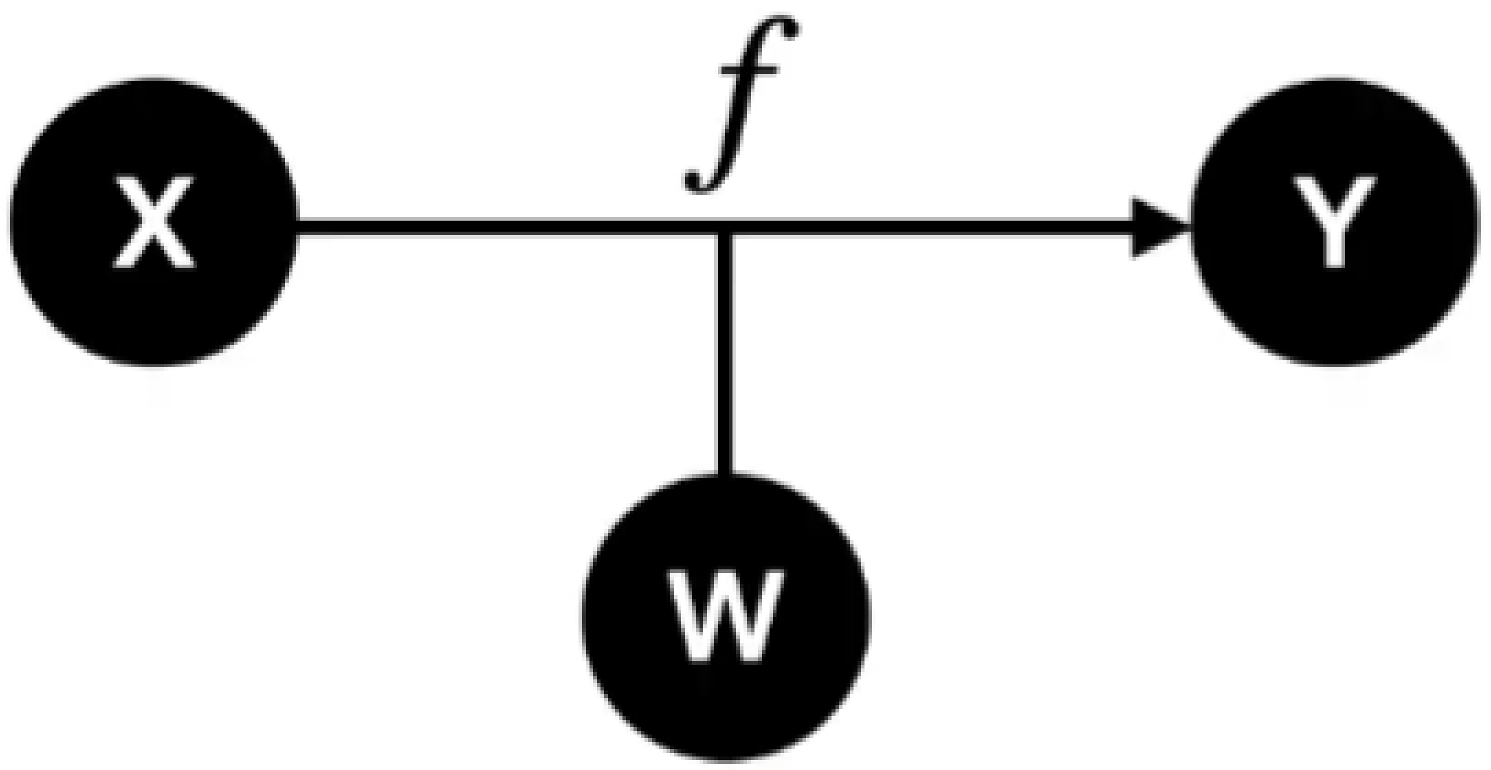
Utility

1. Visualization: Easily <u>see</u> the structure of relationships in the data.

2. Complexity: Captures the complex & hierarchical relationships in the data.

Moderation

Moderation



$$Y = \hat{\beta}_0 + \hat{\beta}_1 X + \hat{\beta}_2 \underbrace{W}_{\text{moderating variable}} + \underbrace{\hat{\beta}_3}_{\text{moderating effect}} XW$$

$\underline{X:}$ Predictor (independent) variable(s)

$\underline{Y:}$ Response (dependent) variable(s)

$\underline{W:}$ Moderator variable(s)

<u>Interpretation:</u>

$\underline{\hat{\beta}_1:}$ Units that Y changes with X

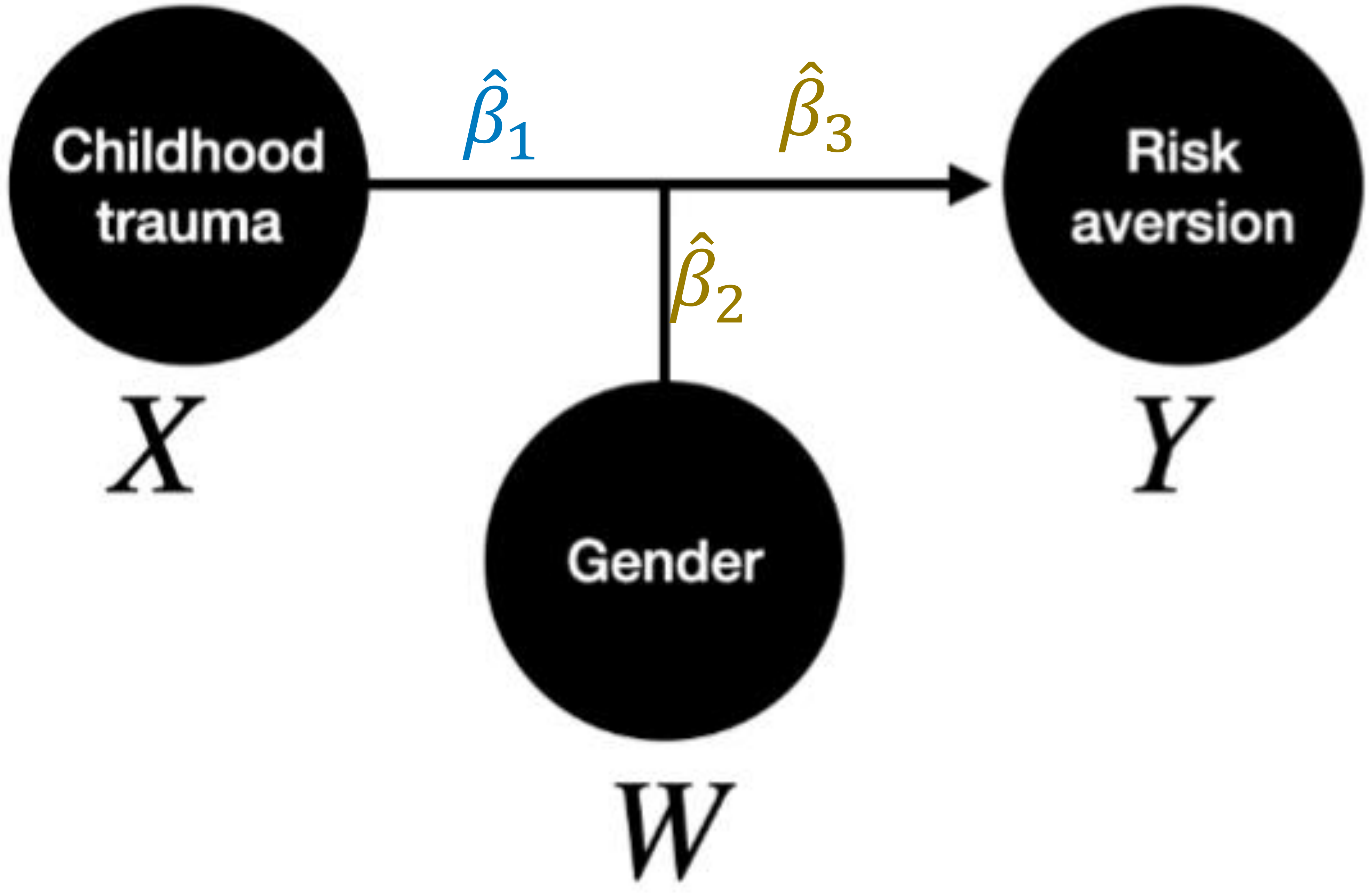
$\underline{\hat{\beta}_2:}$ Units that Y changes with W

$\underline{\hat{\beta}_3:}$ Units that Y changes with X <u>contingent</u> on changes in W

(Preacher & Hayes 2008)

Example: moderation

Q: Is the effect of childhood trauma on risk aversion moderated by gender?



Model:

$$Y = \hat{\beta}_0 + \hat{\beta}_1 X_{CT} + \hat{\beta}_2 W_{gender} + \hat{\beta}_3 X_{CT} W_{gender}$$

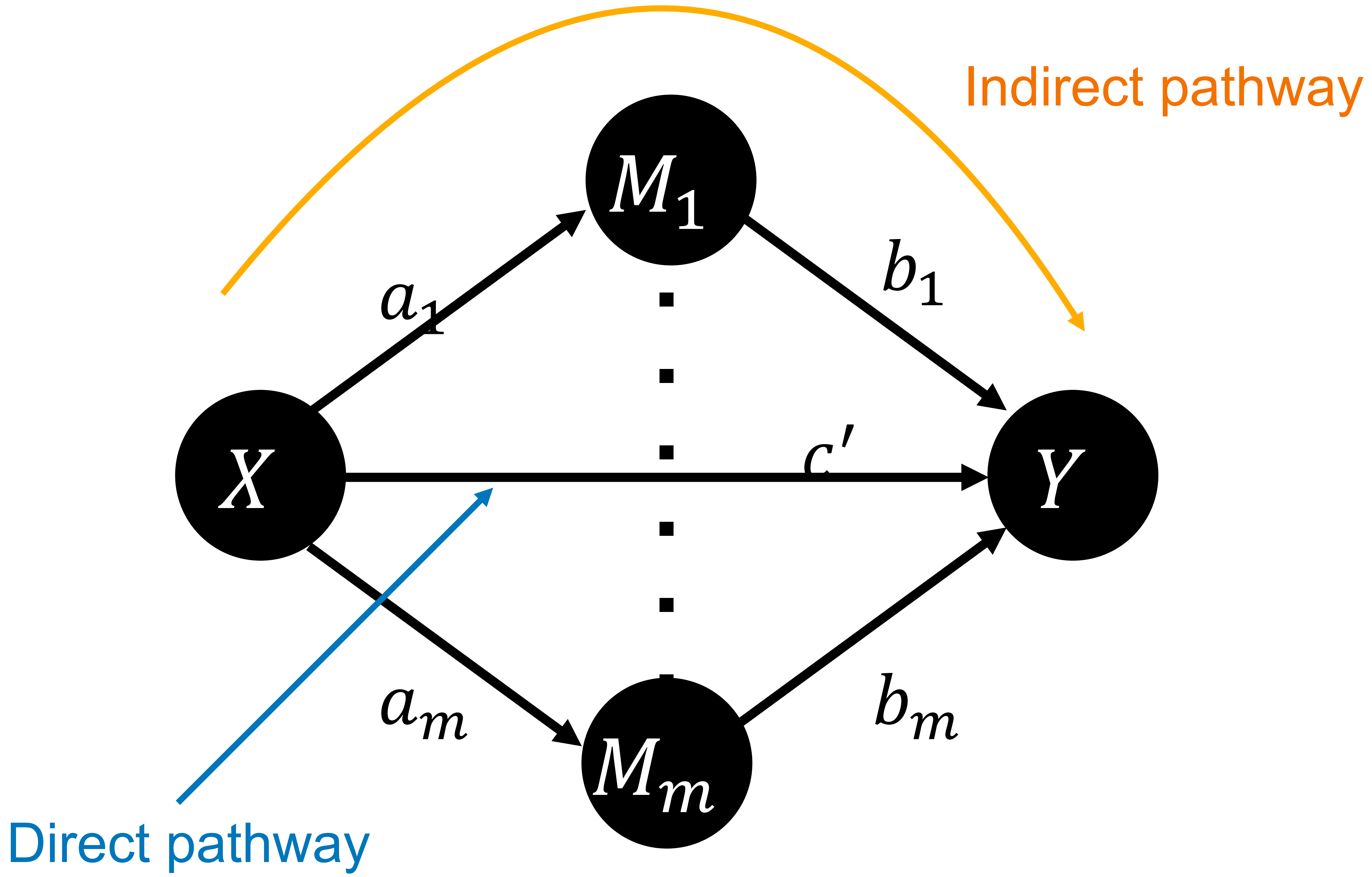
$$Y = \hat{\beta}_0 + (\hat{\beta}_1 + \hat{\beta}_3 W_{gender}) X_{CT} + \hat{\beta}_2 W_{gender}$$

The effect slope of X on Y is: $\hat{\beta}_1 + \hat{\beta}_3 W_{gender}$
(i.e., how much does the slope change with W changes)

(Preacher & Hayes 2008)

Mediation

Mediation models



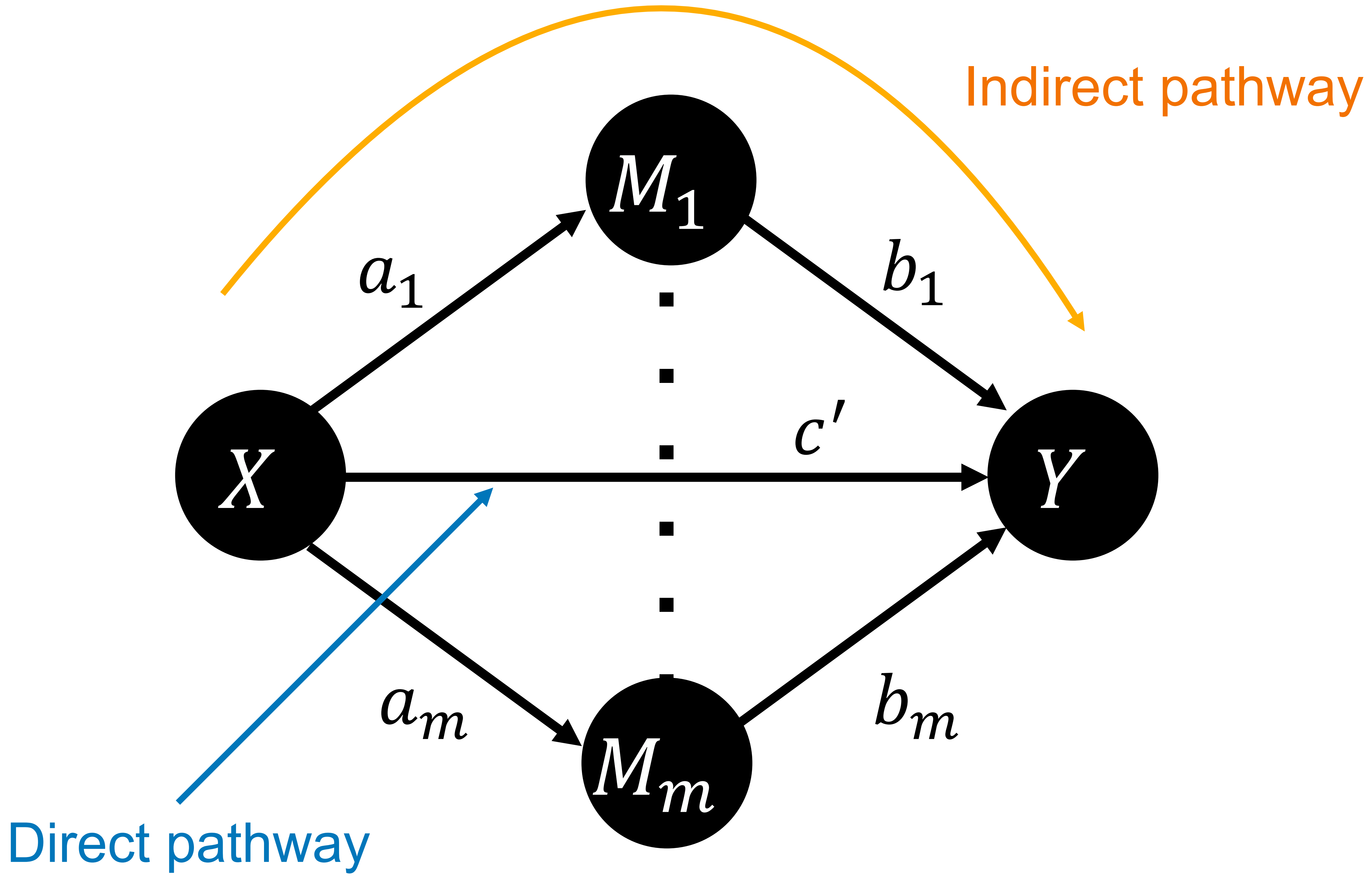
X: Predictor (independent) variable(s).

Y: Response (dependent) variable(s).

M_i: Mediating variable *i*.

(Preacher & Hayes 2008)

Mediation models



<u>Interpretation</u>

a_i: Influence of X on M_i.

b_i: Influence of M_i on Y.

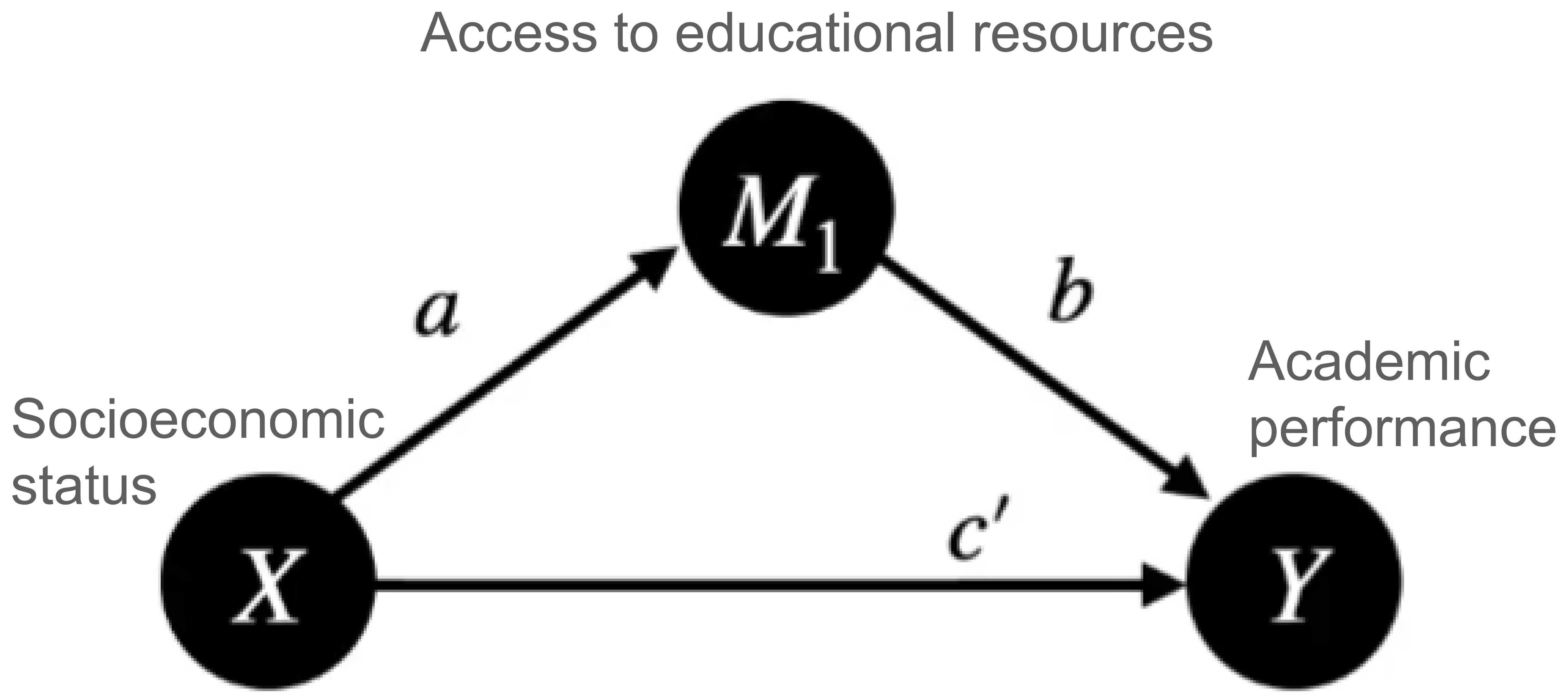
$a_i b_i$: <u>Indirect</u> influence of X on Y via its influence on M_i.

c': <u>Direct</u> influence of X on Y after accounting for all indirect effects (i.e,, all $M_1 \rightarrow M_m$).

c: <u>Total</u> influence of X on Y without accounting for indirect effects.

(Preacher & Hayes 2008)

Estimating mediation effects



$H_0: ab = 0$

Evaluate: Bootstrapping

$$95\%\ \mathrm{CI} = \mathrm{E}[\hat{a}\hat{b}] \pm 1.96\sigma_{bootstrap}$$

Does my CI include 0?

3 regression models

1. $Y = cX$
2. $M = aX$
3. $Y = bM + c'X$

$$\begin{aligned} Y &= bM + c'X \\ &= b(aX) + c'X \\ &= \underbrace{abX}_{indirect} + \underbrace{c'X}_{direct} \end{aligned}$$

(Preacher & Hayes 2008)

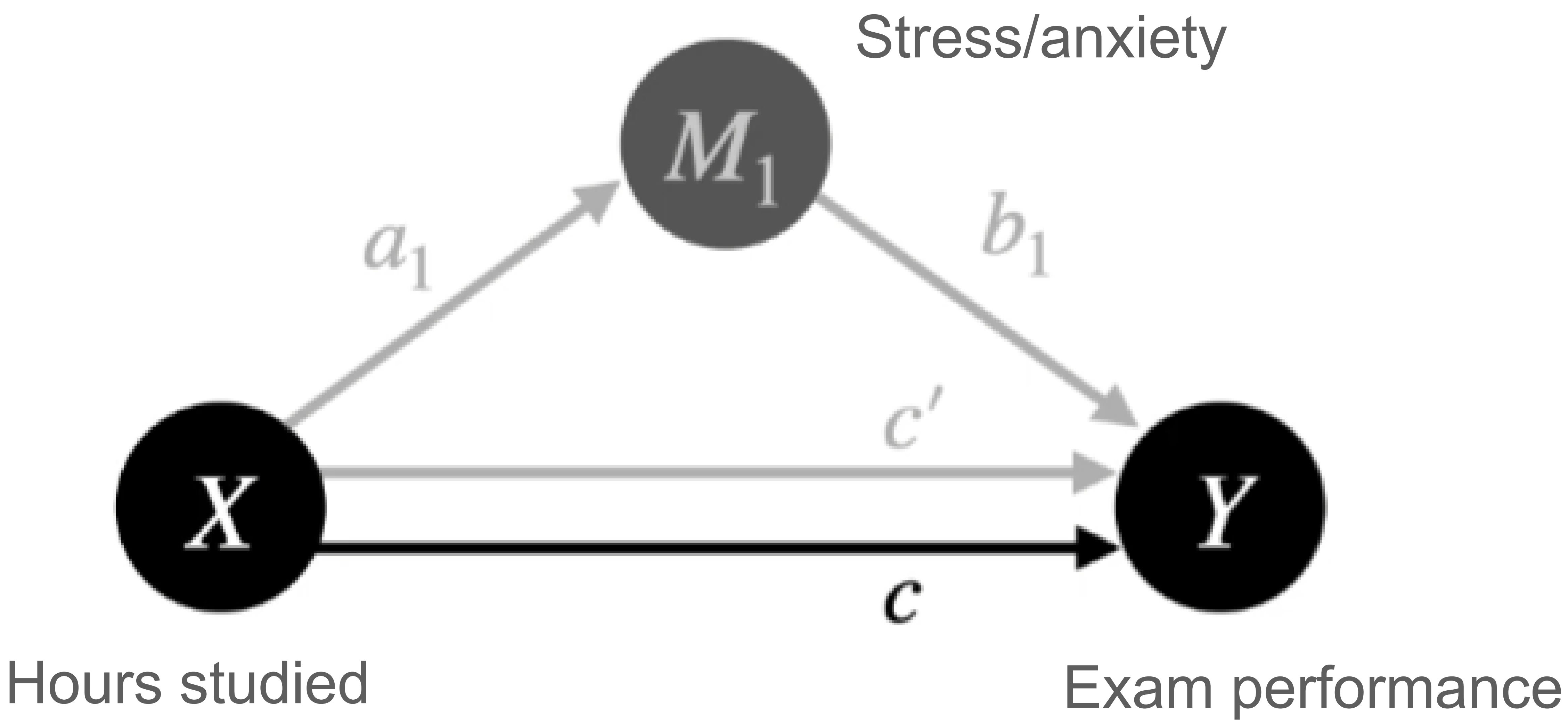
Assumptions

- Both $\hat{a}$ & $\hat{b}$ edges must be $\neq 0$ for the indirect pathway to be evaluated (i.e., cannot infer indirect pathway effects if only one link is non-zero).
    - That is, (X is related to M) & (M is related to Y)
- Very sensitive to low statistical power ($\downarrow power = \uparrow false\ positive\ rate$).

(Preacher & Hayes 2008)

Finding hidden relationships

Sometimes indirect pathways can hide total (c) pathway effects.



$$Y = cX$$
$$= b(aX) + c'X$$
$$= \underbrace{(ab + c')X}_{c}$$

<u>Hidden total path</u>

$$c = (ab + c') = 0$$

$$ab = -2$$
$$c' = +2$$

When direct and indirect pathways have equally opposing influences they cancel each other out.

(Preacher & Hayes 2008)

Multiple mediator models

Q: Is the effect of childhood trauma on risk aversion mediated by parental income, psychiatric risk, and social network size?

X: childhood trauma

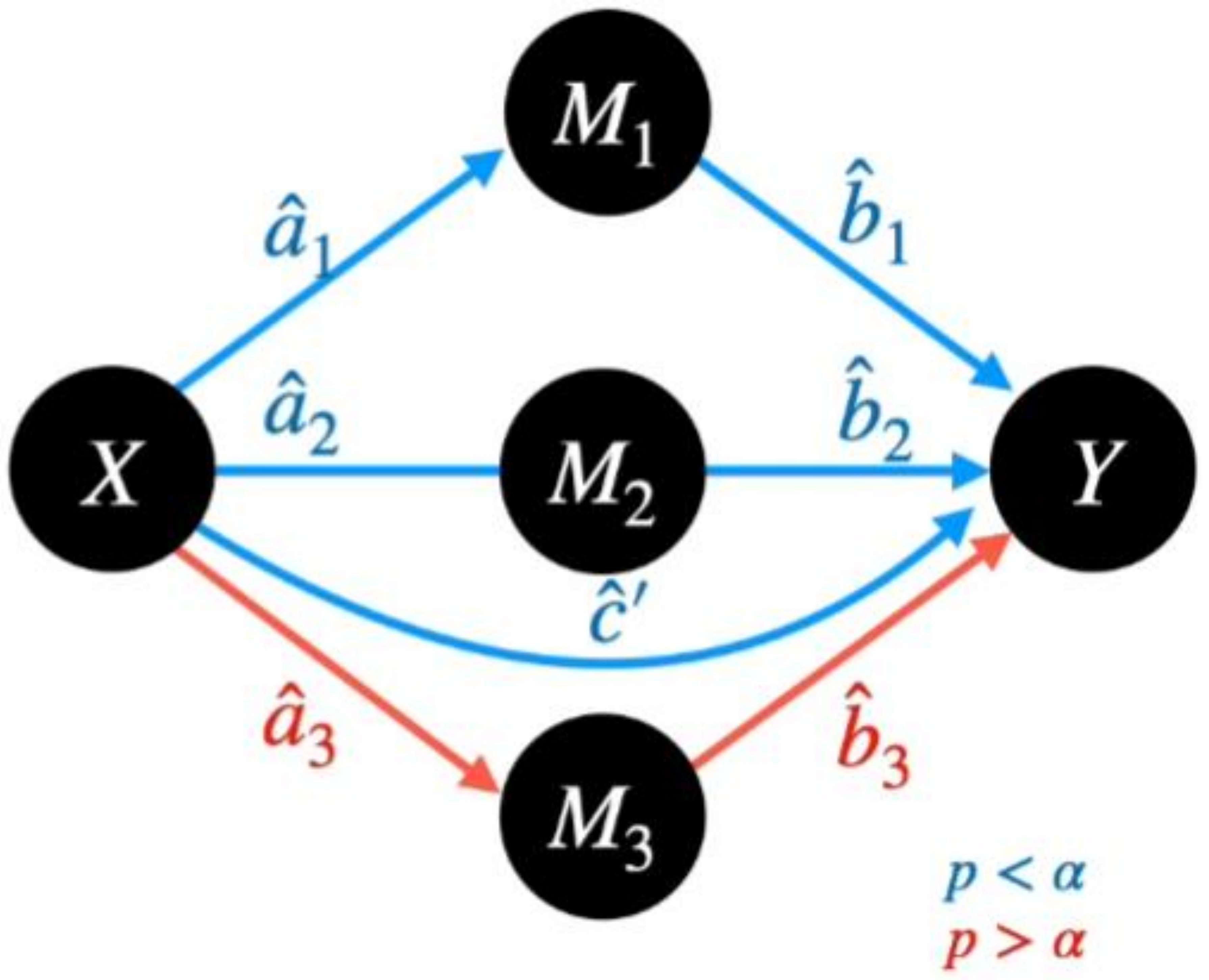
Y: risk aversion

M_1: parental income

M_2: psychiatric risk

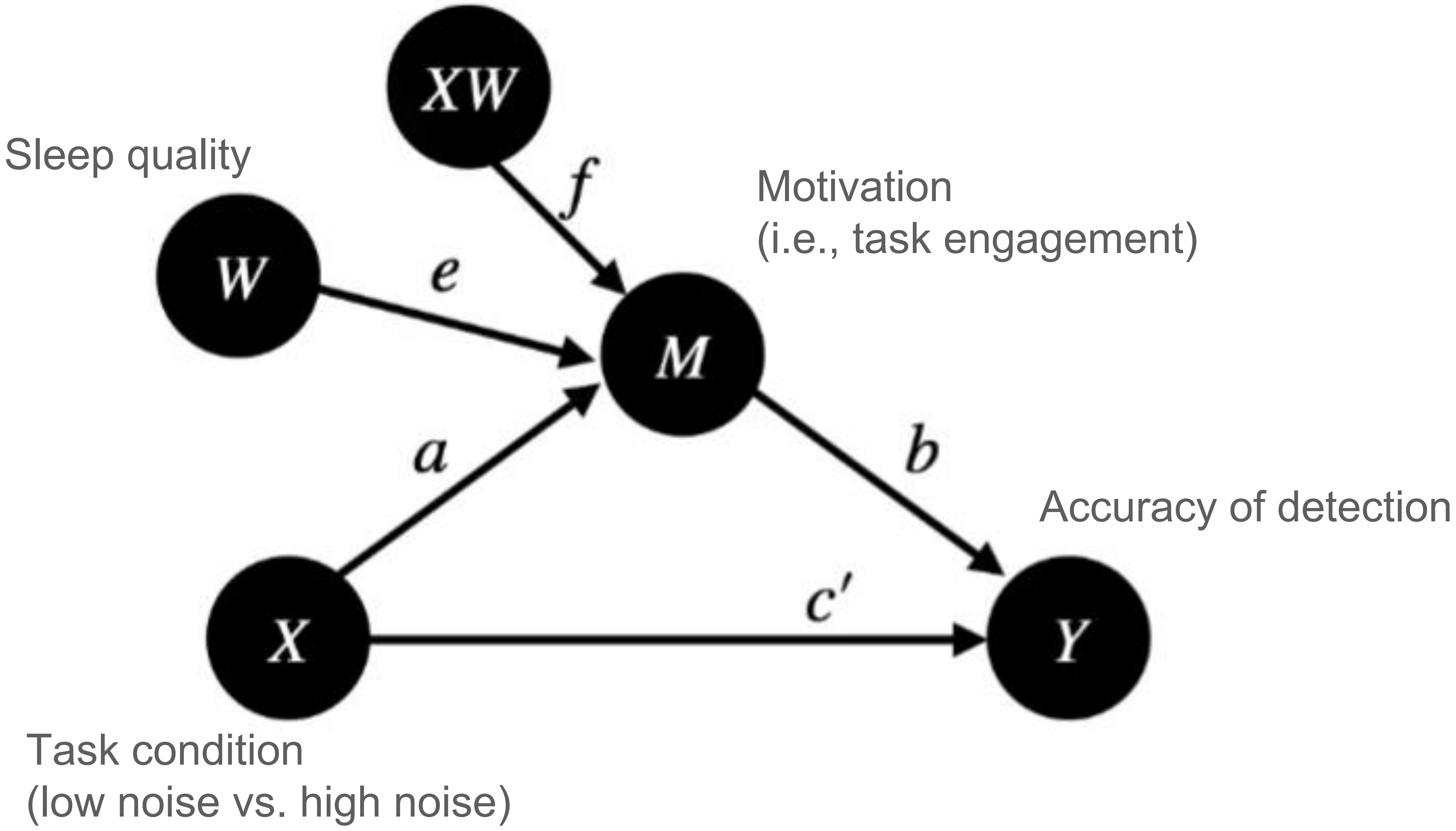
M_3: social network size

<u>Model:</u> $Y = \sum_{i=1}^{p} \hat{a}_i \hat{b}_i X_i + \hat{c}' X$



(Preacher & Hayes 2008)

Moderated mediation models



Does W moderate the indirect relationship between X and Y via M?

Moderated mediation

$$M = aX + eW + fXW$$

a, e: main effects

f: interaction

Full Model

$$Y = \hat{b}M + \hat{c}'X$$
$$= b(aX + eW + fXW) + c'X$$

(Preacher & Hayes 2008)

Take home message

- Representing relations as graphs provides an intuitive understanding of complex relationships.
- Moderation and mediation models allow for capturing relationships beyond first-order associations, even revealing hidden relationships in your data.